Manipulatives by Lesson — Reveal Math, Grade 6 (Units 2–9)

Hands-on and virtual manipulatives matched to each lesson in the Updated Reveal Math slide decks. Each pairing targets the lesson's core concept — the manipulative builds the idea the slides then formalize, so it belongs in the Explore/Warm-Up portion, before the worked examples.
Virtual alternatives: when physical sets are short, Polypad (polypad.amplify.com), Didax Virtual Manipulatives (didax.com/math/virtual-manipulatives.html), and Toy Theater (toytheater.com/category/teacher-tools/virtual-manipulatives) have free browser versions of nearly everything below — algebra tiles, fraction bars, pattern blocks, geoboards, number lines, and base-ten blocks — usable on student devices with no login.

Core kit — what to gather once

Fourteen items cover all 54 lessons. Ordered roughly by how many lessons each one serves.

Manipulative	Lessons served	Why it earns its spot
Linking / Unifix cubes	2-2, 2-5, 2-8, 2-9, 3-1, 6-3, 6-4, 7-3, 7-6, 9-1, 9-3	The single highest-leverage item: data towers, mean leveling, exponents, patterns.
Two-color counters	2-9, 3-1, 3-5, 7-1	Ratios and integers; red/yellow doubles as negative/positive.
Clothesline number line + card sets	2-4, 2-5, 2-6, 7-1, 7-2, 7-4, 8-4	One length of rope and index cards covers the entire number-line strand.
Coordinate grid mat (or painter's tape floor grid)	3-4, 7-5, 7-6, 7-7, 9-2	Every graphing lesson; tape on the floor is a free version.
Algebra tiles (or cups-and-counters)	6-5, 6-6, 6-8, 8-1, 8-2, 8-3	Expressions and equations strand; cups + counters are a free substitute.
Fraction tiles / bars	4-2, 6-1, 6-2	Fraction division and percent-fraction connections.
Centimeter / unit cubes	5-5, 5-6	Volume by filling and counting layers.
Geoboards	5-1, 5-2, 5-4, 7-7	Area derivations and coordinate polygons.
Pattern blocks	5-3, 5-4, 6-2	Trapezoid area and pattern-block fractions.
Base-ten blocks	2-7, 4-1	Decimal division and the hundred-grid meaning of percent.
Hundred grids (printed)	4-1, 4-2, 4-3, 4-4	The core percent model; printable, so effectively free.
Pan balance	8-1, 8-2, 8-3	The equation-solving model; a coat-hanger balance works.
Paper strips (tape diagrams, double number lines, percent bars, nets)	3-2, 3-5, 4-3, 4-4, 4-5, 5-6, 5-7, 5-8	Cardstock + scissors covers the entire ratio/percent strip family and all nets.
Measurement set (rulers in/cm, cup–gallon set, meter stick, scale)	3-6, 3-7	Unit conversion lessons; borrowable from a science classroom.

Unit 2 — Statistics and Data Displays

Lesson	Manipulative(s)	How it helps
2-1 Understand Statistical Questions	Sticky notes; number cubes	Each student answers a survey question on a sticky note and posts it on a whiteboard number line — variability (or the lack of it) becomes visible instantly, which is the whole point of a statistical question.
2-2 Represent and Describe Data in a Histogram	Sticky notes; linking cubes	Move the sticky-note data into labeled interval bins, or stack linking-cube towers per bin — students physically build the bars before drawing them.
2-3 Describe the Data Using the Median	Index cards with data values	Students order the cards, then peel one card off each end simultaneously until the middle remains — the definition of median as a position, not a computation.
2-4 Represent and Describe Data in a Box Plot	Clothesline number line; index cards; string	Hang data cards on the line, find median and quartiles by the peeling routine, then stretch string to mark the box — the five-number summary comes from the data, not a recipe.
2-5 Describe Data by Range and Interquartile Range	Clothesline number line; linking cubes	Measure the physical distance between hung cards for range and IQR; compare two data sets' spreads side by side with cube trains.
2-6 Median and Outliers / Comparing Distributions	Index cards; clothesline number line	Same card-peeling routine as 2-3, now with messier sets (even counts, repeated values, outliers) so the procedure gets stress-tested.
2-7 Divide Decimals Using an Algorithm	Base-ten blocks; grid paper	Let the flat = 1, rod = 0.1, cube = 0.01; sharing blocks into equal groups shows why the algorithm's regrouping steps work before students go symbolic.
2-8 Describe Data Using the Mean	Linking / Unifix cubes	The classic leveling-off model: build a tower per data value, then move cubes until all towers are equal — the mean is what everyone gets when the data is shared fairly.
2-9 Describe Data by Mean Absolute Deviation	Linking cubes on a number line; two-color counters	Mark the mean, then measure each value's distance from it in cubes; red/yellow counters tag above/below. MAD stops being a formula and becomes 'average distance from the mean.'
2-10 Choose Appropriate Measures	Data card sets from earlier lessons	Re-use the physical card sets: add one outlier card and watch the mean-tower shift while the median card barely moves — the argument for choosing measures makes itself.

Unit 3 — Ratios and Rates

Lesson	Manipulative(s)	How it helps
3-1 Understand Ratios	Two-color counters; snap-cube trains	Build 2 red : 3 yellow physically, then copy the group to show 4:6 — a ratio is a relationship you can hold, not a pair of numbers.
3-2 Rates and Unit Rates	Double number line strips; play money	Slide along a paper double number line to find cost per one item; play money makes 'unit price' concrete for the shopping contexts in the deck.
3-3 Equivalent Ratios Using Tables	Color tiles; cuisenaire rods	Build each table row as a tile arrangement — students see the rows are copies of the same unit, which is why the table scales.
3-4 Equivalent Ratios Using Graphs	Coordinate grid mat or taped floor grid	Plot the tile arrangements from 3-3 as points; walking or placing counters on a large grid shows equivalent ratios landing on a straight path through the origin.
3-5 Comparing Ratio Relationships	Two-color counters as 'mix batches'; paper tape-diagram strips	Build both recipes (e.g., lemonade mixes) with counters and compare which is 'stronger' — the comparison argument comes before any computation.
3-6 Converting Within One System	Customary measurement set (cup/pint/quart/gallon), rulers, yard/meter stick	Actually pour, measure, and line up units — conversion factors stop being trivia when students have filled 4 cups into a quart themselves.
3-7 Converting Between Systems	Dual-scale rulers (in/cm); liter bottle next to a quart; balance scale	Side-by-side physical comparison (1 in ≈ 2.5 cm, 1 L vs 1 qt) anchors the approximate conversion factors the lesson uses.

Unit 4 — Percent

Lesson	Manipulative(s)	How it helps
4-1 Understand Percent	Hundred grids (10×10); base-ten flats	Shade or cover cells of a hundred grid — percent literally means ‘per hundred,’ and the grid makes that definition the tool.
4-2 Relate Fractions, Decimals, and Percentages	Hundred grid; fraction tiles with equivalence cards	The same shaded grid reads three ways (43/100, 0.43, 43%); fraction tiles laid on the grid connect benchmark fractions to their percents.
4-3 Estimate the Percent of a Number	Folded percent strips (benchmark 10%, 25%, 50%)	Folding a strip into halves, quarters, and tenths gives students physical benchmarks to estimate against before computing.
4-4 Find and Compare with Percentages	Double number line strips; hundred grid	One line for the quantity, one for 0–100% — sliding along paired lines is the model the deck’s problems assume.
4-5 Determine the Whole Given the Part and Percent	Paper tape-diagram / percent-bar strips; cuisenaire rods	If a rod is 25%, how long is 100%? Iterating the part to rebuild the whole makes the ‘reverse’ percent problem forward again.

Unit 5 — Area, Volume, and Surface Area

Lesson	Manipulative(s)	How it helps
5-1 Area of Parallelograms and Rhombuses	Grid paper + scissors; geoboards	Cut the triangle off one end of a paper parallelogram and slide it to the other — the rectangle formula transfers, and students proved it themselves.
5-2 Area of Triangles	Geoboards; congruent paper triangles	Two copies of any triangle snap together into a parallelogram — that is the entire ½bh argument, done in students' hands.
5-3 Area of Trapezoids	Pattern blocks (red trapezoid); paper cutting	Two trapezoid blocks make a parallelogram; the block everyone already knows carries the derivation.
5-4 Apply Area Concepts to Solve Problems	Pattern blocks; color tiles; tangrams	Compose and decompose composite figures physically before splitting them on paper.
5-5 Volume of Rectangular Prisms	Centimeter / unit cubes and small boxes	Fill a box, count layers — volume as layers × layer-count is the model; essential when the deck moves to fractional edge lengths.
5-6 Three-Dimensional Figures in Two Dimensions	Foldable cardstock nets (Polydron/Geofix if available); isometric dot paper	Folding a net into its solid and flattening it again is the exact 2D↔3D translation the lesson practices.
5-7 Surface Area of Prisms	Deconstructed cereal boxes; square tiles	Cutting a real box open along its edges produces the net, and 'surface area = area of the net' becomes obvious.
5-8 Surface Area of Pyramids	Foldable pyramid nets	Same move as 5-7: unfold, identify the base and triangular faces, sum the areas you can see.

Unit 6 — Expressions

Lesson	Manipulative(s)	How it helps
6-1 Division Expressions with Fractions and Whole Numbers	Fraction tiles / bars; paper strips	‘How many ¼s fit in 3?’ answered by laying tiles — division as measurement, before the invert-and-multiply shortcut.
6-2 Division Expressions with Fractions and Mixed Numbers	Fraction bars; pattern blocks (hexagon = 1 whole)	Pattern-block fractions (halves, thirds, sixths) let students physically fit divisors into mixed-number amounts.
6-3 Explore Numerical Expressions with Exponents	Linking cubes; grid paper	Build 4^2 as a square and 4^3 as a cube — the names ‘squared’ and ‘cubed’ stop being arbitrary.
6-4 Write and Evaluate Numerical Expressions with Exponents	Linking cubes; color tiles	Same builds, now paired with writing the expression each structure represents.
6-5 Write and Evaluate Algebraic Expressions	Algebra tiles; cups-and-counters	A cup holding an unknown number of counters IS the variable; evaluating means filling the cup and counting.
6-6 Identify Equivalent Algebraic Expressions	Algebra tiles	Combine like tiles: 3x + 2 + x rearranges to 4x + 2 physically, so equivalence is ‘same pile, different arrangement.’
6-7 Find Factors and Multiples	Color tiles; hundred chart with counters	Every rectangle you can build from 24 tiles is a factor pair; skip-counting counters on a hundred chart shows common multiples landing on the same squares.
6-8 Generate Equivalent Expressions	Algebra tiles in an area model	Build 3(x + 2) as three rows of an x-tile and two unit tiles — the distributive property is just reading the same rectangle two ways.

Unit 7 — Integers, Rational Numbers, and the Coordinate Plane

Lesson	Manipulative(s)	How it helps
7-1 Explore Integers and Their Opposites	Two-color counters; clothesline number line	Red = negative, yellow = positive; a zero pair in the hand plus opposites hung symmetrically about 0 on the line.
7-2 Rational Numbers and Their Opposites on the Number Line	Clothesline number line with fraction/decimal cards	Students place −1.5 and −⅗ and defend the position — placement arguments are the lesson.
7-3 Understand Absolute Value	Number line + string or cube train	Measure the distance from zero with string: |−4| and |4| use the same length of string, which is the definition.
7-4 Compare and Order Integers and Rational Numbers	Clothesline number line with mixed card sets	Ordering by physically re-hanging cards forces the 'further left = less' reasoning that trips students up with negatives.
7-5 Rational Numbers on the Coordinate Plane	Coordinate grid mat / taped floor grid	Students walk or place markers at all four quadrants — the sign pattern of the quadrants is learned with their feet.
7-6 Distance on the Coordinate Plane	Grid mat + string or cube chains	Lay cubes between two points that share a coordinate and count — then connect the count to subtracting coordinates and absolute value.
7-7 Polygons on the Coordinate Plane	Geoboards with labeled axes; rubber bands	Stretch the polygon from its vertex coordinates, then find side lengths by counting pegs — plotting and measuring in one tool.

Unit 8 — Equations and Inequalities

Lesson	Manipulative(s)	How it helps
8-1 Understand Equations and Their Solutions	Pan balance with cubes; cups-and-counters	A solution is whatever keeps the balance level — testing values physically is exactly what the lesson's 'is it a solution?' items ask.
8-2 Solve Equations Using Addition or Subtraction	Pan balance; algebra tiles	Remove the same number of cubes from both pans and it stays level — the 'do the same to both sides' rule earned, not announced.
8-3 Solve Equations Using Multiplication or Division	Algebra tiles; balance with equal groups	Split both sides into equal groups; one group per x-tile shows why dividing both sides isolates the variable.
8-4 Write and Represent Inequalities	Number line with open/closed circle chips and arrow cards	Placing the right chip (open vs. closed) and arrow makes the graphing conventions a decision, not a memorized mark.
8-5 Understand Inequalities and Their Solutions	Number line + counters for test values	Drop a counter on a value, test it in the inequality, leave it if it works — the solution set literally accumulates on the line.

Unit 9 — Relationships Between Two Variables

Lesson	Manipulative(s)	How it helps
9-1 Explore Relationships Between Two Variables	Function machine (box + input/output cards); growing patterns in cubes	Feed inputs into the box, record outputs, hunt the rule; cube-train growing patterns give the same relationship in a buildable form.
9-2 Analyze Graphs of Relationships	Coordinate grid mat	Plot the function-machine table from 9-1 and watch the rule become a line; ‘walking the graph’ on a floor grid connects steepness to rate.
9-3 Write Equations to Represent Relationships	Input/output card tables; cube patterns	Figure n of a cube pattern forces the jump from ‘add 3 each time’ to ‘y = 3x + 1’ — the equation is the shortcut students want.
9-4 Apply Two-Variable Relationships	Play money and counters for cost/earning scenarios	Act out the deck’s real-world contexts (cost per item, earnings per hour) so the independent/dependent roles are physical before they are symbolic.